Model Development Document

Complaint → Regulation Classifier (CMPL-REG-24)
First line of defense — Model Development

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LLM: nim (meta/llama-3.1-8b-instruct)
Data: CFPB Consumer Complaint Database (real, redacted narratives)
Dataset: 4000 rows · regulatory rate 0.9663
Repo: reg-agents · model id: CMPL-REG-24

1 · Executive summary & methodology

Executive Summary

This two-stage complaint-to-regulation classification model is designed to accurately categorize consumer complaints into specific regulatory categories. The model's primary objective is to provide a robust and reliable framework for identifying potential regulatory issues, thereby enabling timely interventions and mitigating potential risks. The model's performance is evaluated using a comprehensive set of metrics, including precision-recall area under the curve (PR-AUC), area under the receiver operating characteristic curve (ROC-AUC), and family accuracy.

The model's two-stage architecture is a deliberate design choice, driven by the need to balance model performance and computational efficiency. The first stage employs a cheap high-recall gate, utilizing a TF-IDF classifier, to quickly filter out a significant proportion of complaints that are unlikely to be relevant to specific regulatory categories. This stage is designed to be computationally inexpensive, allowing for the efficient processing of large volumes of complaints. The second stage, on the other hand, employs a more expensive and sophisticated retrieval-augmented generation pipeline (rag_llm), which is used to refine the predictions made by the first stage and provide more accurate and nuanced categorizations.

Methodology Narrative

The model development process began with the selection of a suitable dataset, comprising 4000 complaints from the CFPB Consumer Complaint Database. The dataset was split into training, validation, and test sets, with 3040 complaints used for training, 380 for validation, and 380 for testing. The regulatory rate, which represents the proportion of complaints that are relevant to specific regulatory categories, was found to be 0.9663.

The first stage of the model employed a bake-off between two TF-IDF classifiers: logistic regression and XGBoost. The logistic regression model, with a l1 regularization parameter of 2.0, achieved a PR-AUC of 0.9957 on the validation set and a PR-AUC of 0.9933 on the test set. The XGBoost model, with a maximum depth of 6 and a learning rate of 0.1, achieved a PR-AUC of 0.9901 on the validation set and a PR-AUC of 0.9882 on the test set. The logistic regression model was ultimately selected as the champion, due to its superior performance on the PR-AUC metric.

The second stage of the model employed a retrieval-augmented generation pipeline (rag_llm), which was used to refine the predictions made by the first stage. The rag_llm model achieved an accuracy of 0.3913 on the test set, a family accuracy of 0.5652, and a macro F1 score of 0.3467. The model's performance varied across different regulatory categories, with some categories exhibiting high recall rates (e.g., FCRA_PERMISSIBLE_PURPOSE, FDCPA_COMMUNICATION, and RESP_A_SERVICING) and others exhibiting low recall rates (e.g., GLBA_PRIVACY, LOAN_SERVICING, and REG_CC_FUNDS).

Known Weaknesses

One of the primary weaknesses of the model is the use of weak labels, which can lead to biased and inaccurate categorizations. The model's performance on the weak-label reference is a concern, as it may not accurately reflect the true regulatory categories of the complaints. Additionally, the model's reliance on a single dataset may limit its generalizability to other complaint datasets.

Monitoring and Guardrail Design

To mitigate the risks associated with the model's weaknesses, a comprehensive monitoring and guardrail design has been implemented. The model's performance will be continuously monitored, and any significant changes in its behavior will be investigated and addressed. Additionally, a set of

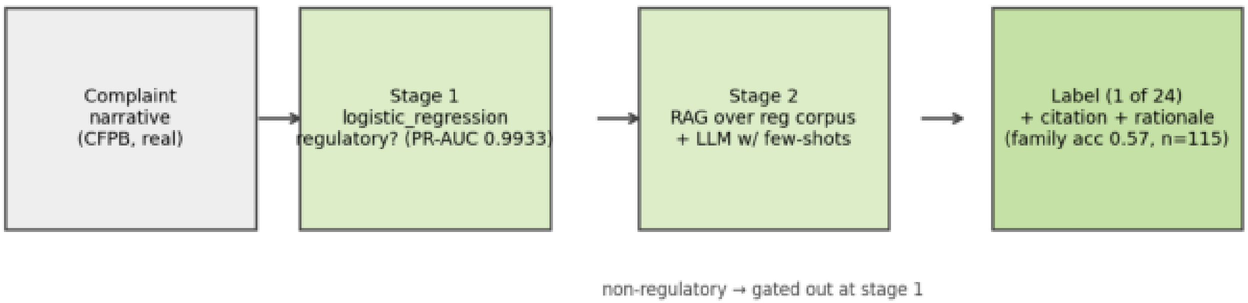
1 · Executive summary & methodology (cont.)

guardrails has been established to prevent the model from making inaccurate or biased categorizations. These guardrails include:

* A minimum accuracy threshold of 0.5 for the second stage of the model
* A maximum family accuracy threshold of 0.7 for the second stage of the model
* A minimum recall rate of 0.2 for each regulatory category

These guardrails will help to ensure that the model's predictions are accurate and unbiased, and that any potential risks associated with its weaknesses are mitigated.

2 · Architecture



Two-stage pipeline: binary gate, then RAG+LLM labeling with citations.

2 · Architecture (detail)

Stage 1 (binary gate): TF-IDF (1-2 grams, 30k features) into a logistic-regression vs XGBoost bake-off; champion selected on PR-AUC. Upgrade path: fine-tuned BERT-class encoder (NeMo Framework) served via Triton, same interface.

Stage 2 (multi-class): retrieval-augmented generation over the regulation/policy corpus (NeMo Retriever embeddings, FAISS locally / cuVS-Milvus on GPU) + LLM reasoning (NIM Llama-3.1-8B) with 8 curated few-shot examples and the 24-category taxonomy in-prompt. Output is strict JSON: label, confidence, rationale, and the cited source document, which the UI renders alongside the excerpt. A keyword scorer provides a deterministic no-LLM fallback; stage-1 gates non-regulatory complaints so the LLM is only invoked when needed.

Deployment: complaint MCP server (tools: classify_complaint, sample_complaints, get_model_metrics) + Complaint A2A agent; wired into the Streamlit UI and observable via the existing Prometheus/Grafana and OpenTelemetry stack.

3 · Data & curation

Source: real, redacted consumer complaint narratives from the public CFPB Consumer Complaint Database (4000 curated rows; 200 reserved as a 5% scoring holdout for the batch-ingestion layer before any modeling split, then 3040 train / 380 validation / 380 test; regulatory rate 0.9663).

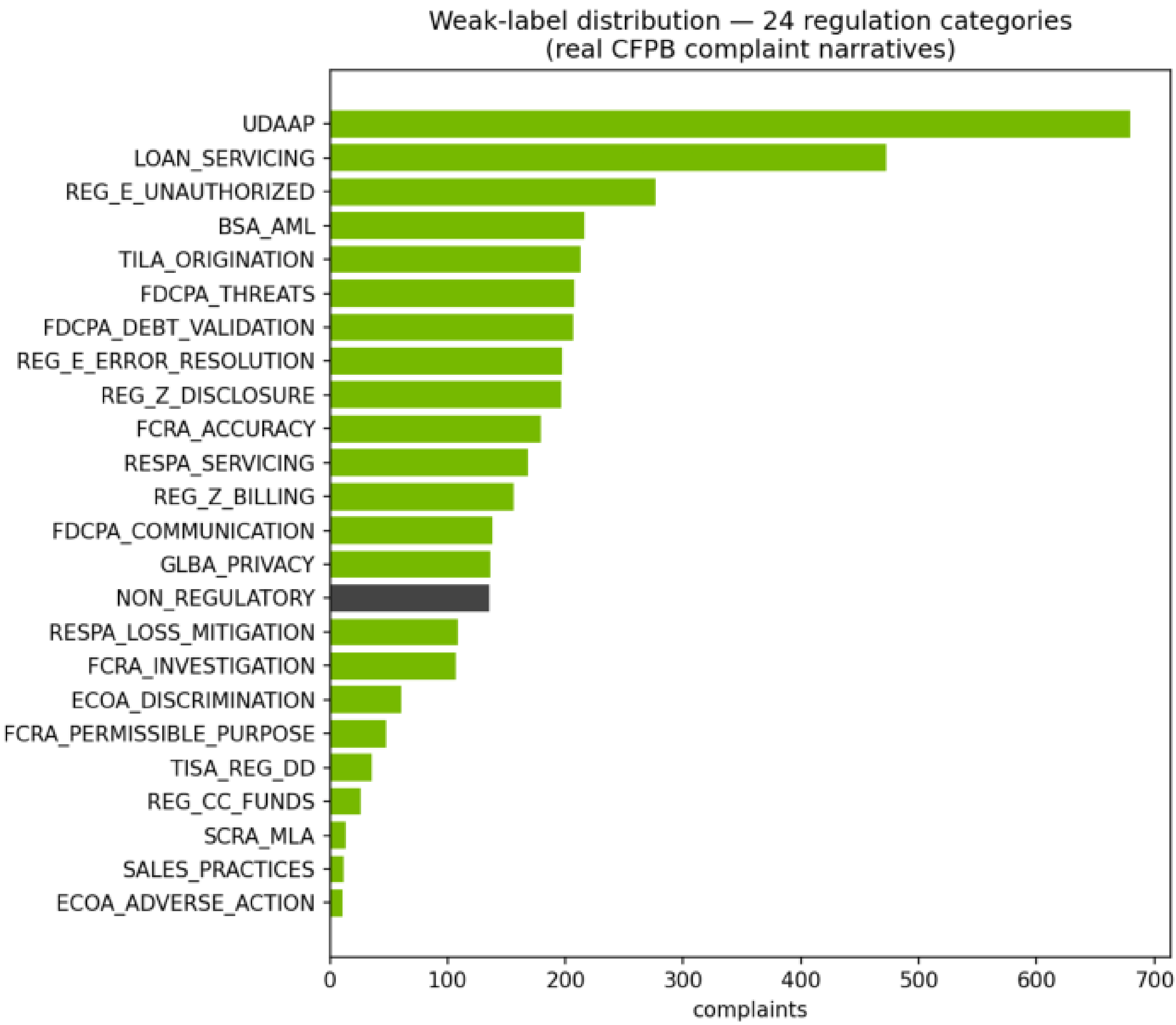
Curation (scripts/fetch_cfpb_complaints.py) mirrors NVIDIA NeMo Data Curator stages: length filtering (ScoreFilter), exact deduplication (ExactDuplicates), near-deduplication (FuzzyDuplicates/MinHash analog), PII verification (CFPB pre-masks PII as XXXX; PiiModifier analog), and per-issue balanced sampling. At corpus scale each stage maps directly to the GPU-accelerated Curator module.

Ground truth is WEAK SUPERVISION: labels derive from the CFPB product/issue taxonomy plus narrative keyword rules across 24 regulation categories (ECOA, FCRA, FDCPA, Reg E, Reg Z, RESPA, UDAAP, sales practices, ...). This is the standard bootstrap before human-adjudicated labels exist and is flagged as a limitation in the validation report.

3 · Dataset summary

property	value
source	CFPB Consumer Complaint Database (public, PII-redacted)
curated rows	4,000
scoring holdout (reserved first)	200 (5%, stratified — fed only through the ingestion layer)
train / val / test split	3,040 / 380 / 380 (stratified 80/10/10 of the remaining 95%)
regulatory rate	96.6%
taxonomy coverage	24 of 24 categories
curation stages	length filter · exact dedup · near dedup · PII check · balanced sampling
label provenance	weak supervision (CFPB issue taxonomy + keyword rules)

3 · Label distribution



Weak-label distribution over the 24 regulation categories.

3.1 · Regulation taxonomy (with dataset support)

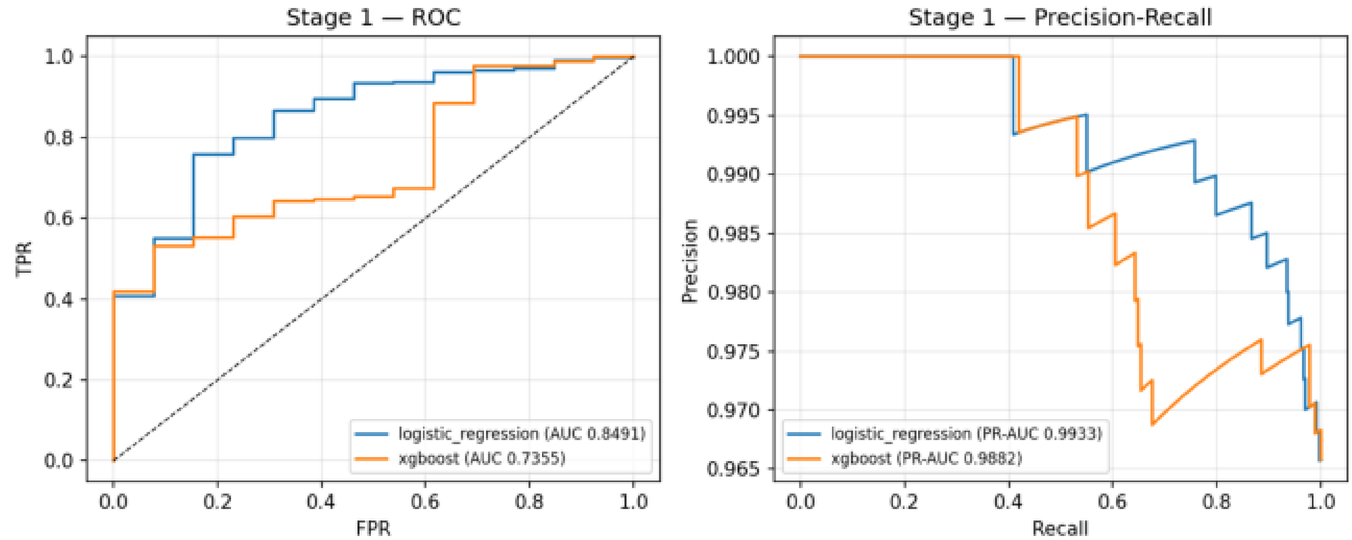
code	regulation / category	n in dataset
FCRA_ACCURACY	FCRA — Accuracy of Reported Informatio	179
FCRA_INVESTIGATION	FCRA — Reinvestigation of Disputes	107
FCRA_PERMISSIBLE_PURPOSE	FCRA — Permissible Purpose / Improper U	48
FDCPA_DEBT_VALIDATION	FDCPA — Debt Validation / Not Owed	207
FDCPA_COMMUNICATION	FDCPA — Communication Tactics	138
FDCPA_THREATS	FDCPA — False Statements or Threats	208
REG_E_UNAUTHORIZED	Reg E / EFTA — Unauthorized Transfers	276
REG_E_ERROR_RESOLUTION	Reg E / EFTA — Error Resolution	197
REG_Z_BILLING	Reg Z / FCBA — Billing Error Disputes	156
REG_Z_DISCLOSURE	Reg Z / TILA — Fees, Interest & Disclosure	196
TILA_ORIGINATION	TILA — Credit Origination & Underwriting Disc	213
ECOA_DISCRIMINATION	ECOA / Reg B — Credit Discrimination	61
ECOA_ADVERSE_ACTION	ECOA / Reg B — Adverse Action Notices	11
RESPA_SERVICING	RESPA — Mortgage Servicing & Escrow	168
RESPA_LOSS_MITIGATION	RESPA — Loss Mitigation & Foreclosure	109
TISA_REG_DD	TISA / Reg DD — Deposit Account Disclosu	36
REG_CC_FUNDS	Reg CC — Funds Availability	26
BSA_AML	BSA/AML — Account Freezes & Closures	216
GLBA_PRIVACY	GLBA / Privacy — Information Sharing & Safeg	136
UDAAP	UDAAP — Unfair, Deceptive, or Abusive Ac	680
SALES_PRACTICES	Sales Practices — Unauthorized Accounts/Pro	12
SCRA_MLA	SCRA / MLA — Servicemember Protection	13
LOAN_SERVICING	Loan Servicing (Auto/Student/Personal) — UDAAP & S	472
NON_REGULATORY	Non-Regulatory — General Service	135

4 · Stage-1 bake-off leaderboard

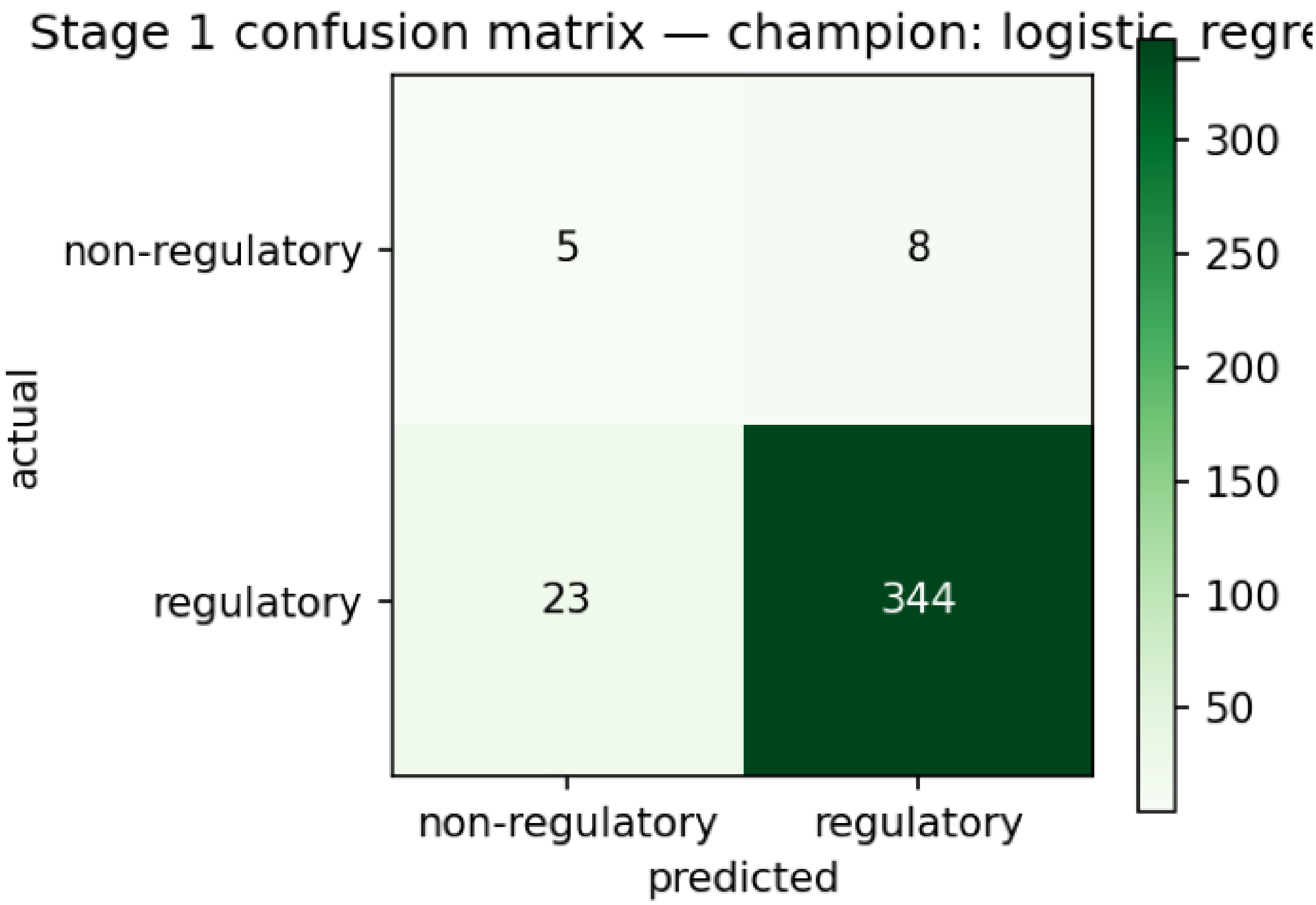
model	params	val_pr_au	threshold	pr_auc	train_roc_a	roc_auc	train_test_g	f1	precision	recall	accuracy
logistic_regres	l1, C=2.0	0.9957	0.639	0.9933	0.9966	0.8491	0.1475	0.9569	0.9773	0.9373	0.9184
xgboost	n_estimators=300, max_c	0.9901	0.844	0.9882	1.0	0.7355	0.2645	0.9453	0.974	0.9183	0.8974

Champion: logistic_regression - selected on validation PR-AUC; other columns are one-shot test metrics.

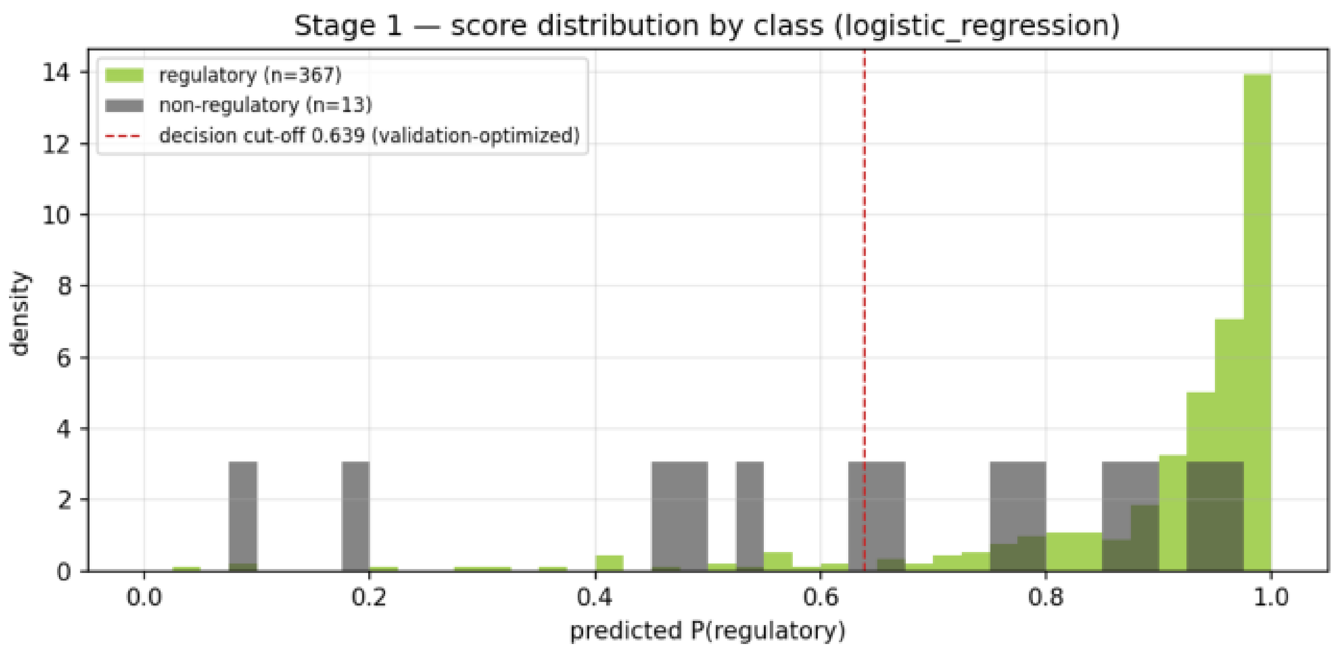
4 · Stage-1 ROC / PR curves



4 · Stage-1 confusion matrix

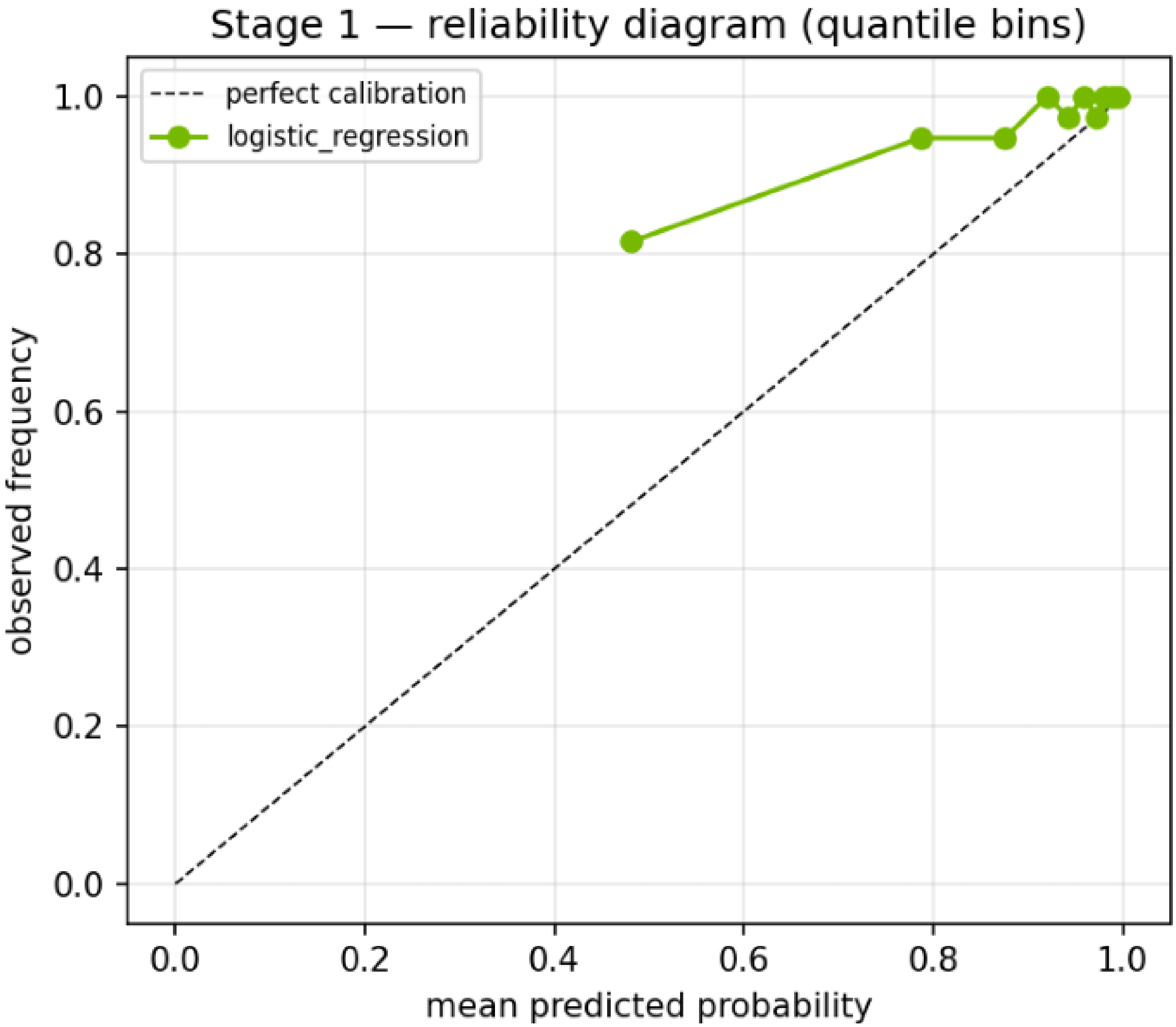


4.1 · Score separation



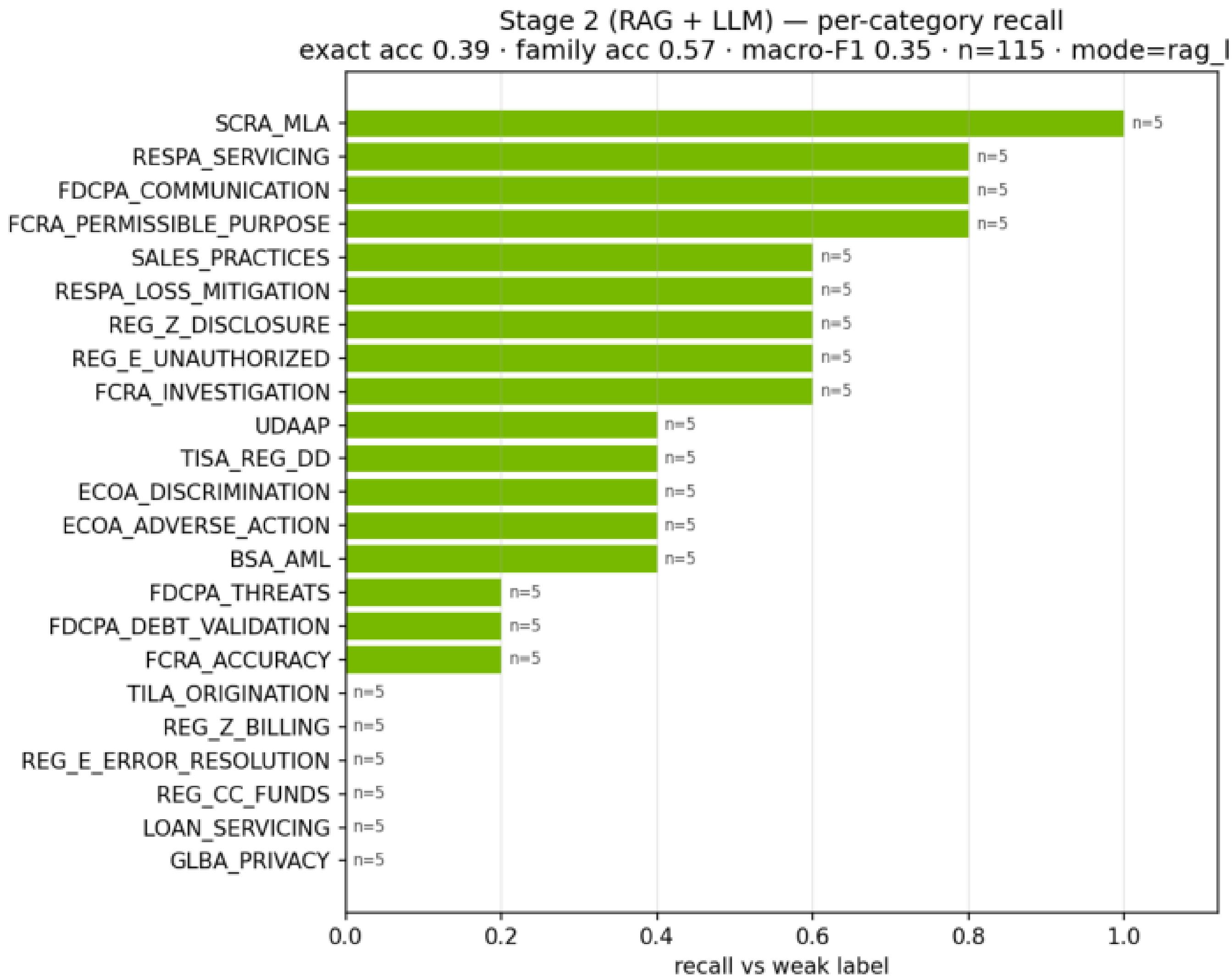
Class separation at the validation-optimized cut-off 0.639 (default 0.5 not used).

4.1 · Calibration (reliability diagram)



Whether predicted probabilities can be read as probabilities.

5 · Stage-2 per-category recall



Exact acc 0.3913 · family acc 0.5652 · macro-F1 0.3467 · n=115.

5 · Stage-2 per-category detail

category	support	recall
BSA_AML	5	0.4
ECOA_ADVERSE_ACTION	5	0.4
ECOA_DISCRIMINATION	5	0.4
FCRA_ACCURACY	5	0.2
FCRA_INVESTIGATION	5	0.6
FCRA_PERMISSIBLE_PURPOSE	5	0.8
FDCPA_COMMUNICATION	5	0.8
FDCPA_DEBT_VALIDATION	5	0.2
FDCPA_THREATS	5	0.2
GLBA_PRIVACY	5	0.0
LOAN_SERVICING	5	0.0
REG_CC_FUNDS	5	0.0
REG_E_ERROR_RESOLUTION	5	0.0
REG_E_UNAUTHORIZED	5	0.6
REG_Z_BILLING	5	0.0
REG_Z_DISCLOSURE	5	0.6
RESPA_LOSS_MITIGATION	5	0.6
RESPA_SERVICING	5	0.8
SALES_PRACTICES	5	0.6
SCRA_MLA	5	1.0
TILA_ORIGINATION	5	0.0
TISA_REG_DD	5	0.4
UDAAP	5	0.4

Recall vs weak labels on the stratified evaluation sample.

6 · Guardrail inventory

guardrail	mechanism	surfaced as
Taxonomy whitelist	LLM label must be one of the 24 codes; else re	complaint_classifications_total{label}
Strict-JSON parsing	malformed LLM output -> deterministic keyword	mode=fallback counter (alertable spike)
Stage-1 gating	non-regulatory volume never reaches the L	cost + attack-surface control
Retrieval grounding	answer must cite a retrieved corpus excer	citation attached to every prediction
Provider portability	OpenAI <-> NIM via config; keyword mode if b	mode label per classification

Sign-off

Prepared and reviewed under the bank's model risk management framework (SR 11-7 / OCC 2011-12).

Author, development document: Senior Model Developer & Quantitative Lead — PhD (Econometrics), 20 years in model development across credit, capital planning, fraud, and NLP.

Author, validation report: Senior Independent Model Validator — PhD (Econometrics), 20 years in second-line validation; independent of the development team per SR 11-7 organizational-independence requirements.